

电子科技大学
格拉斯哥学院
Glasgow College, UESTC

UESTC 3002: Artificial Intelligence & Machine Learning

Fall Semester 2024-2025

Lab 3 - Week 15

Generating Dataset

Submission Due Date: By the end of your lab session.

Student's Chinese Name	
Student's English Name	
Student's UESTC ID	
Student's UoG ID	

Generating Dataset: Label and Format

Introduction

In this lab, students will gain hands-on experience in collecting and labeling images of litter and recyclable materials, which will form the basis for a supervised learning task. The objective is to create a dataset that will help train and test machine learning models for image classification. Throughout this exercise, you will learn the importance of well-labeled data in supervised learning, where each input image (such as a plastic bottle or paper waste) is paired with a target label indicating its category. This task not only reinforces theoretical concepts discussed in lectures but also demonstrates how machine learning can be applied to real-world challenges, such as waste management and recycling. By the end of the lab, you will have constructed a valuable dataset, which will be used in future labs to develop and evaluate image classification models.

Install Labelme

LabelMe is an open-source annotation tool primarily used for image labeling. It helps users annotate images for tasks like object detection, segmentation, and classification. Users can manually draw polygons around objects in an image and assign labels to those regions, often for training machine learning models, especially in computer vision tasks.

We encourage you to explore more information on its official GitHub link. Here is a short instruction of installation

```
#prerequest is pyqt
pip install pyqt5
# install label me
pip install labelme
```

if you are using the anaconda, the actual execution file is located in:

```
YOUR_ANACONDA_INSTALL_FOLDER/envs/YOUR_ENV_NAME/Scripts/lablem(.exe)
```

Download the material from the moodle

The folder should should contains the following content.

```
d----           25/10/2024    03:13                example
d----           25/10/2024    03:13                utils
-a---           25/10/2024    03:16             69 label.txt
```

```
-a---          25/10/2024     03:13           6815 labelme2coco.py
-a---          25/10/2024     03:16           2705 README.md
```

Data collection

The categories of garbages

- **Recyclable Paper**
  Examples: newspapers, cardboard, books, office paper.
- **Recyclable Plastic**
  Examples: plastic bottles, plastic packaging, plastic toys.
- **Recyclable Metal**
  Examples: tin cans, aluminum cans, scrap copper, scrap iron.
- **Recyclable Glass**
  Examples: glass bottles, glass jars, broken glass.
- **Food Waste**
  Examples: food scraps, vegetable peels, bones, eggshells.
- **Hazardous Waste**
  Examples: used batteries, expired medications, fluorescent tubes, chemicals.
- **Electronic Waste (E-waste)**
  Examples: old phones, computers, chargers, electronic components.

Task 1: Image Collection

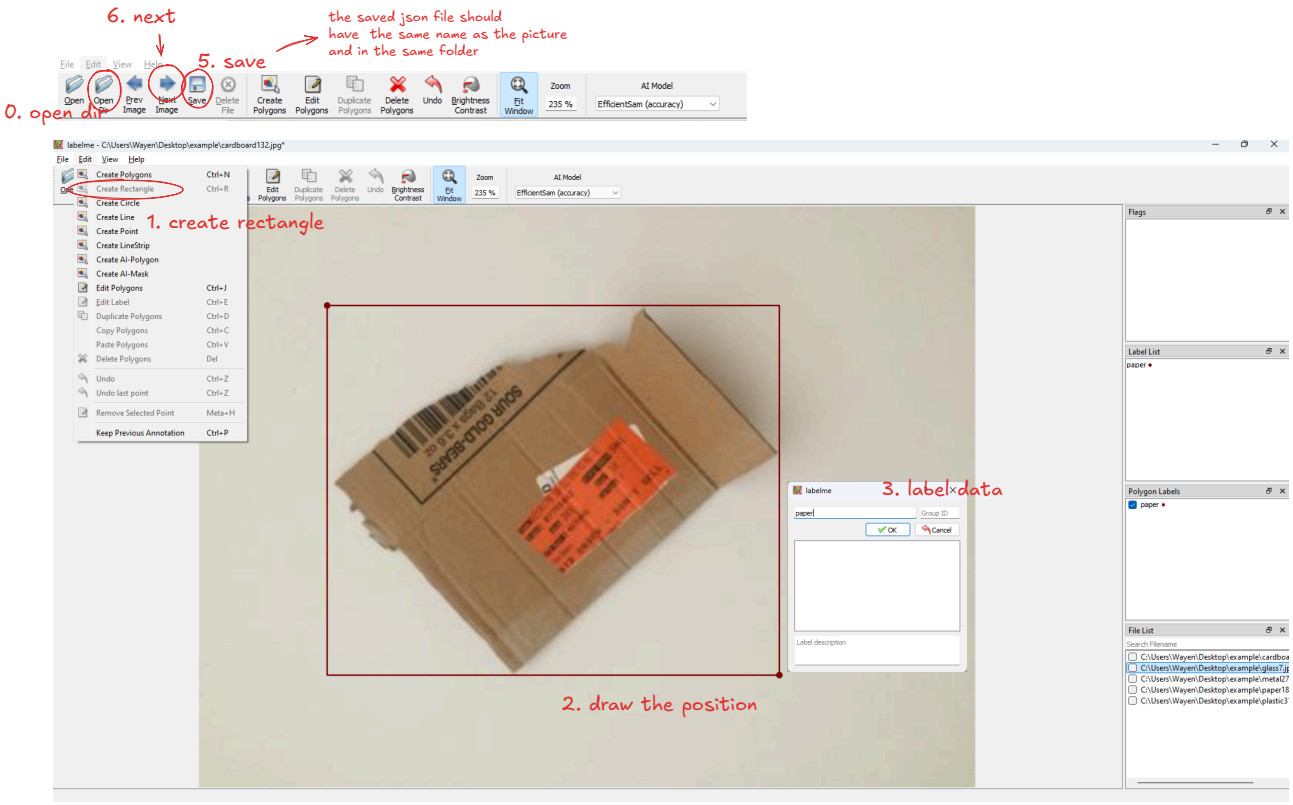
Your task is to collect **FIVE** unique images from each category of litter and recyclable materials. Hence, each student will collect and submit 35 images in total. To ensure variety, no more than two images should be taken of the same object, and you should avoid duplicating the same image. Each image must capture a distinct example of the object under observation. This will help create a diverse and representative dataset for our supervised learning task. Be mindful to ensure that each image is unique, as this is essential for building an effective and accurate machine learning model.

Labeling

Labeling the collected samples is a crucial step in supervised learning, as it directly impacts the performance of the machine learning model. In this section, you will learn how to assign appropriate labels to each image you have collected, using one of the predefined labels. This step ensures that each image is correctly categorized, forming a well-structured dataset that can be used for training and testing in future tasks. Remember, accurate labeling is key to building a successful model.

Here is an illustration of datalabelling using Lableme:

1. Put all your images inside a single folder
2. Open the labelme, and open the folder
3. Label each image with an **RECTANGLE**, an illustration is shown as follows:



The files in the data folder should looks like this:

```
-rw-r--r--@ 1    10K 25 Oct 00:46 cardboard132.jpg
-rw-r--r--@ 1   498B 25 Oct 00:46 cardboard132.json
-rw-r--r--@ 1    10K 25 Oct 00:46 glass7.jpg
-rw-r--r--@ 1   490B 25 Oct 00:46 glass7.json
-rw-r--r--@ 1    21K 25 Oct 00:46 metal27.jpg
-rw-r--r--@ 1   492B 25 Oct 00:46 metal27.json
-rw-r--r--@ 1    25K 25 Oct 00:46 paper18.jpg
-rw-r--r--@ 1   491B 25 Oct 00:46 paper18.json
-rw-r--r--@ 1   9.3K 25 Oct 00:46 plastic31.jpg
-rw-r--r--@ 1   494B 25 Oct 00:46 plastic31.json
```

⚠ Attention

Only JPG image is accepted.
Please rename your image as "<student_id>_<category>_<index>", eg "[redacted]_metal_0", note the index should start from 0.
Besides, the name of the annotated json file should be the same with the original image name (this is default behaviour of Labelme).

Task 2: Labeling the Collected Samples

Your task is to label all the collected image samples accurately. Each label must strictly belong to one of the following categories: metal, plastic, paper, glass, food, hazardous, electronic. Ensure that every image is correctly categorized based on the material or type of item in the image, as this will form the foundation for training your machine learning model.

Convert data to COCO format

By default, the annotation file generated by Labelme has its own format. However, the well-known COCO format is more suitable in context of machine learning and deep learning. Thus we are going to convert it to the COCO data format.
Luckily, the Labelme provided a script to do that.

1. In cmdline, enter the repo folder
2. execute the script by

```
python3 labelme2coco.py /path/to/your/labelled/data
/path/to/output/coco.json --labels label.txt --noviz
```

Submit

Task 3: Submit the Collected Dataset

Once you have finished labeling and converting the data, the next step is to submit your dataset. Each student must rename their dataset files using the following format: [GUID_YourName_Category] (e.g., 5638179_JohnWick_Raw and 5638179_JohnWick_COCO). They should be compressed in zip format. After renaming, submit both dataset to the designated Moodle page by the due date. Make sure all files are correctly labeled and submitted on time to avoid losing marks.